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ITECH3102 – Data Visualization
Week 7

Lab Exercise – Complex data visualizations (2)

ID: _____Name: _____

In previous projects, we learned basic approaches of data visualization and discussed how to visualize complex data. Here visualizing complex data indicates that visualization cannot be easily achieved through **directly using existing fields** from raw data source because these fields don't exist in raw data source. Instead, for visualizing complex data, we may require the use of some more-advanced techniques. For example, we may need to take simple computation on the existing fields or to combine the existing fields or to use other methods in order to make new dimensions or measures that don't exist in raw data source.

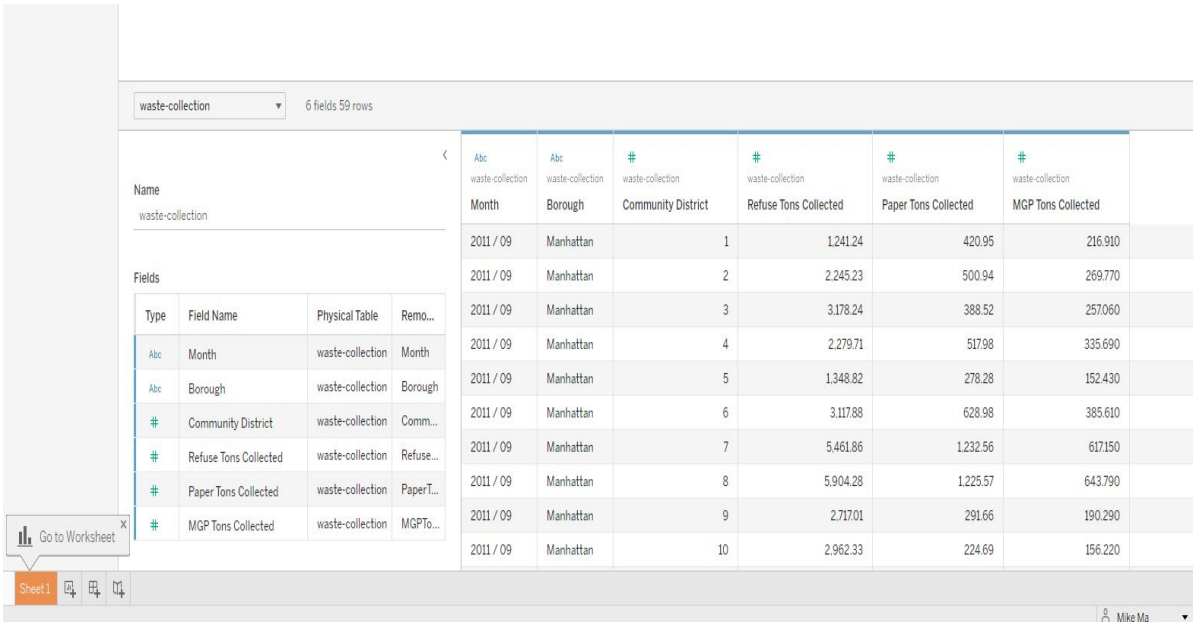
In this project, we apply calculated fields and other techniques to visualize data. For example, for given data, we want to make a comparison in order to answer some questions, such as ratio with comparing two terms and comparison of part to whole, etc.

Project 1: Visualize comparison using calculated fields

Task: The given dataset provides information, such as about the amount of non-recyclable garbage, amount of recyclable paper in a city, etc. The goal of this task is to compare which districts in the city are doing a better job.

1. Download Dataset: waste-collection.xlsx from Week 7 in Moodle

Step 1. Load the data into Tableau and check that your data has been loaded correctly.



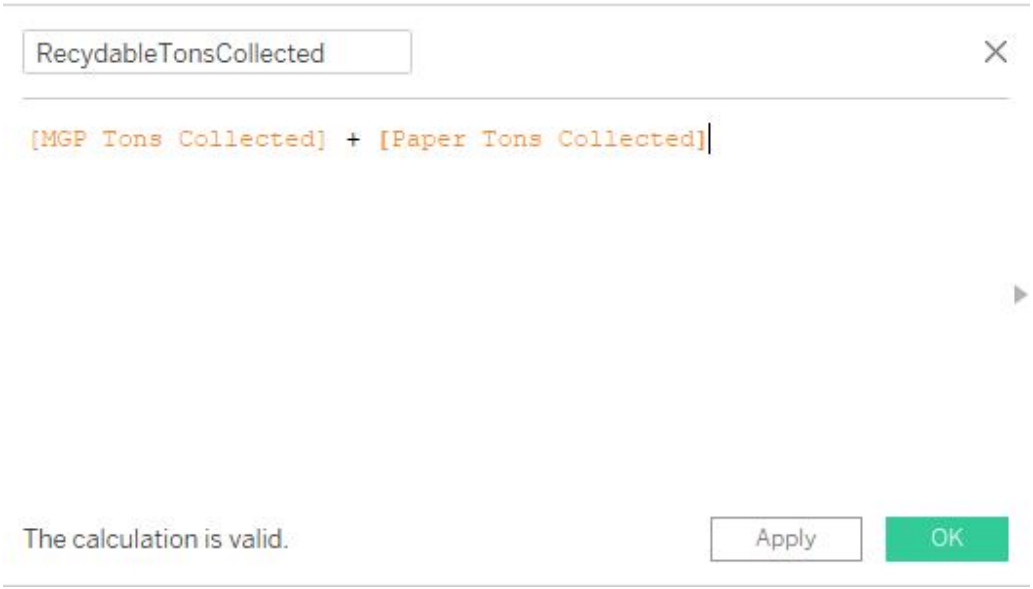
Month	Borough	Community District	Refuse Tons Collected	Paper Tons Collected	MGP Tons Collected
2011 / 09	Manhattan	1	1,241.24	420.95	216.910
2011 / 09	Manhattan	2	2,245.23	500.94	269.770
2011 / 09	Manhattan	3	3,178.24	388.52	257.060
2011 / 09	Manhattan	4	2,279.71	517.98	335.690
2011 / 09	Manhattan	5	1,348.82	278.28	152.430
2011 / 09	Manhattan	6	3,117.88	628.98	385.610
2011 / 09	Manhattan	7	5,461.86	1,232.56	617.150
2011 / 09	Manhattan	8	5,904.28	1,225.57	643.790
2011 / 09	Manhattan	9	2,717.01	291.66	190.290
2011 / 09	Manhattan	10	2,962.33	224.69	156.220

After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

Step 2. Create 2 calculated fields.

We first create a new calculated field called RecydableTonsCollected. Right click any location in Dimensions or Measures area in Data pane and select Create | Create Calculated Field.

Then enter the name of the calculated field and enter code in code editor area to perform the sum calculation as following.



The new calculated field sums the two recyclable fields. After clicking OK button, a new field of RecydableTonsCollected you just created appears in the Measures area.

We then need to create the second calculated field called Recycle to
Refuse Ratio in order to compute the ratio of recyclable material to refuse.

Right click any location in Dimensions or Measures area in Data pane and select Create Calculated Field.

Then enter the name of the calculated field and enter code in code editor area to perform the ratio calculation as following.



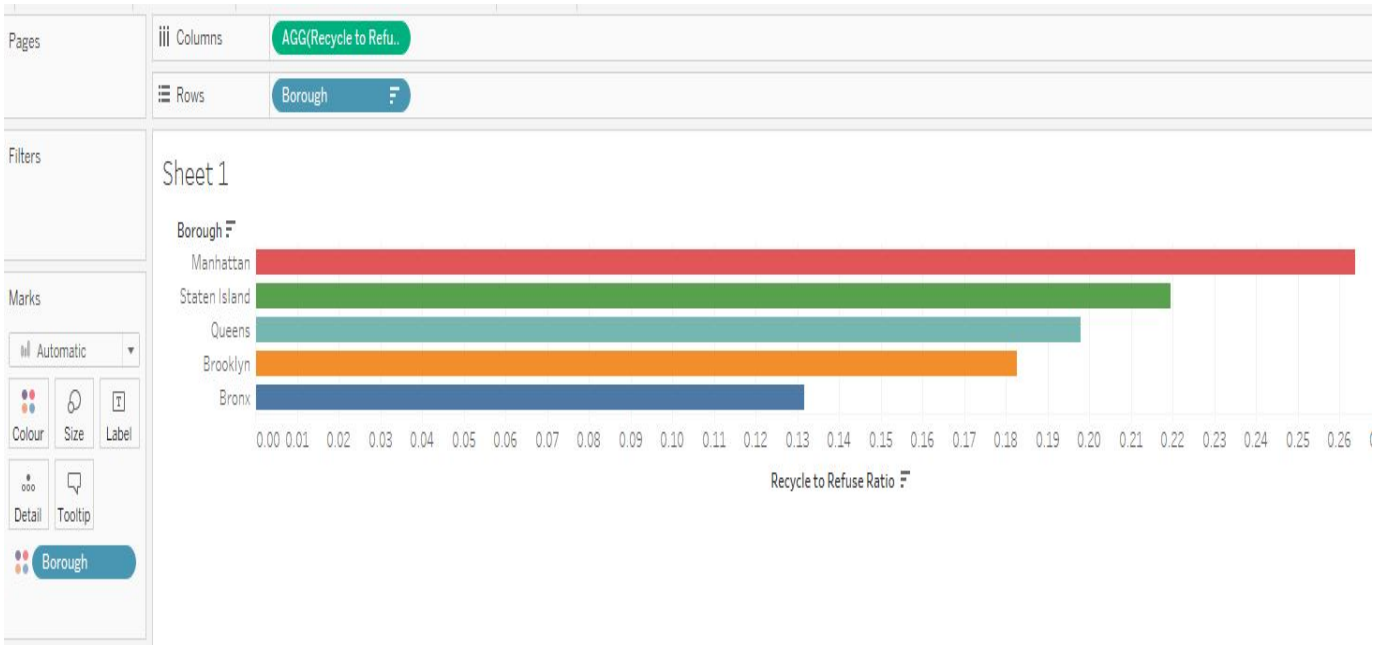
In the second calculated field, we use SUM in the numerator and denominator in brackets because we don't want to **add the individual ratio** of each community district. Here the new ratio represents the quotient of the aggregate amounts.

After clicking OK button, a new field of Recycle to Refuse Ratio you just created appears in the Measures area.



Step 3. Visualize the comparisons at high level (borough)

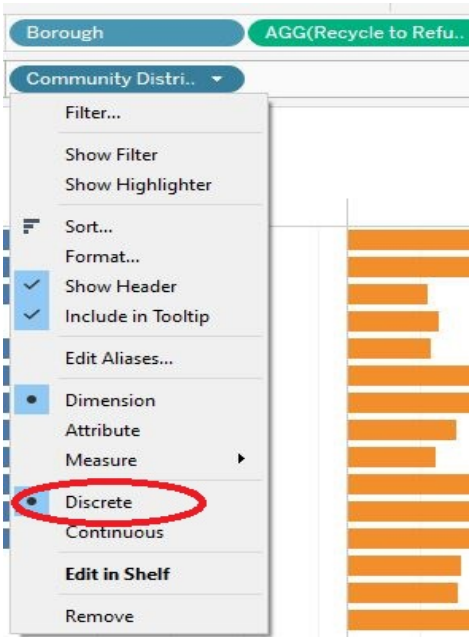
Drag the Borough field from the dimensions area on Data pane to the Rows shelf, and drag the Recycle to Refuse Ratio calculated field to the Columns shelf, then drag Borough to the Color shelf in Marks card, and finally sort the bars in descending order to put the borough with the highest recycle ration at the top.



From visualizing the ratio at the high level for each borough, we can see that Manhattan has the highest ratio of recycle to refuse. However, from visualizing the ratio at the low level for community district, it is not necessarily for each community district in Manhattan has a higher recycle ratio than each community district in the district Bronx. Thus, we are interested in visualizing community district ratios for each borough.

Step 4. Visualize the comparisons at low level (community district)

Drag the Borough from the Rows shelf to the Columns shelf, and drag the Community District field from Measures on Data pane to Rows shelf to create a grid of bar chart. Make sure to select Discrete for Community District.





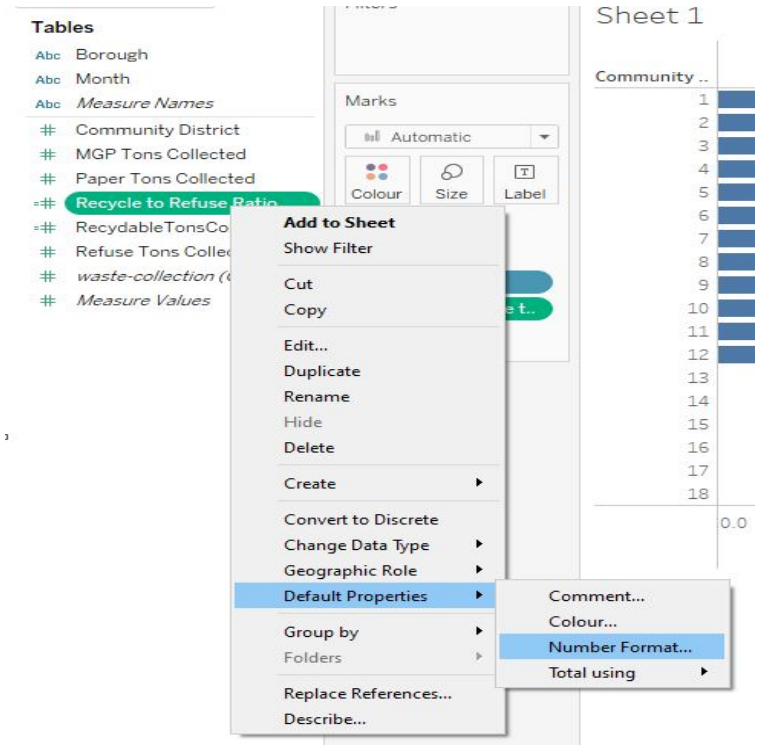
From visualizing community district ratios for each borough, we can see that some community districts in the Bronx have higher ratios than some in Manhattan even if Manhattan community district #1 has the highest ratio overall.

Step 5. Add labels to the bar chart

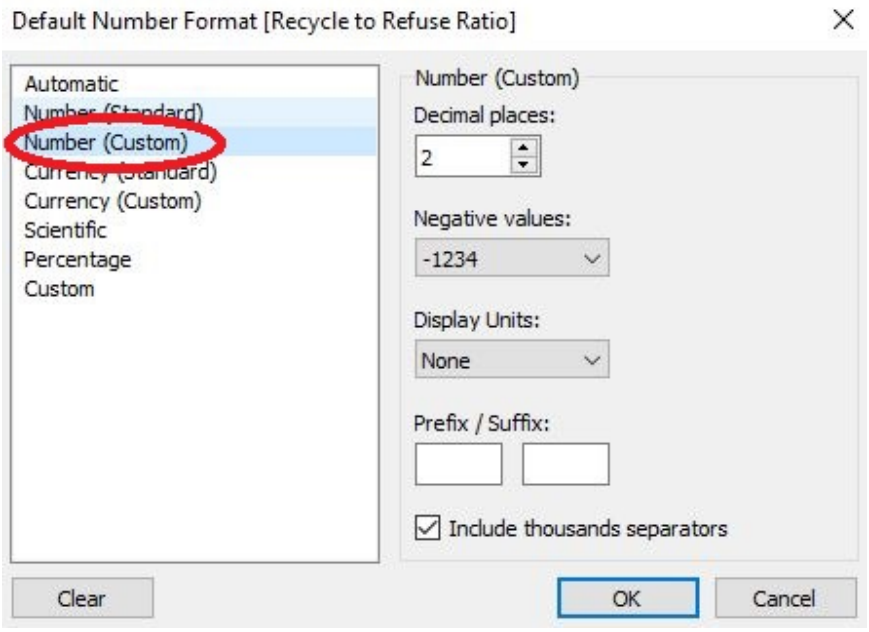
Drag calculated field Recycle to Refuse Ratio from Measure area on Data pane to the Label in Marks card.



We can change the number of digits because there are a lot of decimal places on the labels. Right-clicking Recycle to Refuse Ratio in Measures area, selecting Default Properties, then Number Format.



After selecting Number Format, and we can change the decimal places by choosing Number(custom) in the following box



After clicking Ok, we'll get:



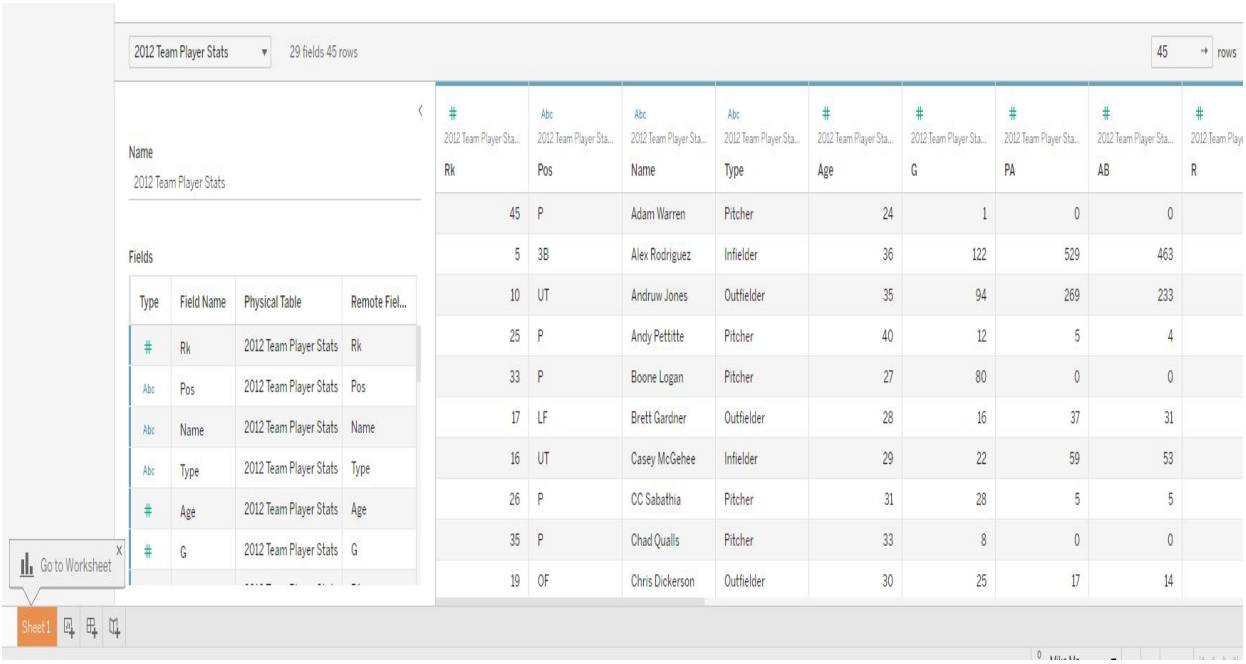
Project 2: Visualize comparison using filters

Task: The given dataset provides statistic information on baseball game's players, including players' names, team batting, team pitching, and batting average, etc. The objective of this task is to compare players' performance.

1. Download Dataset: baseball-player-stats.xlsx from Week 7 in Moodle

Step 1. Load the data into Tableau

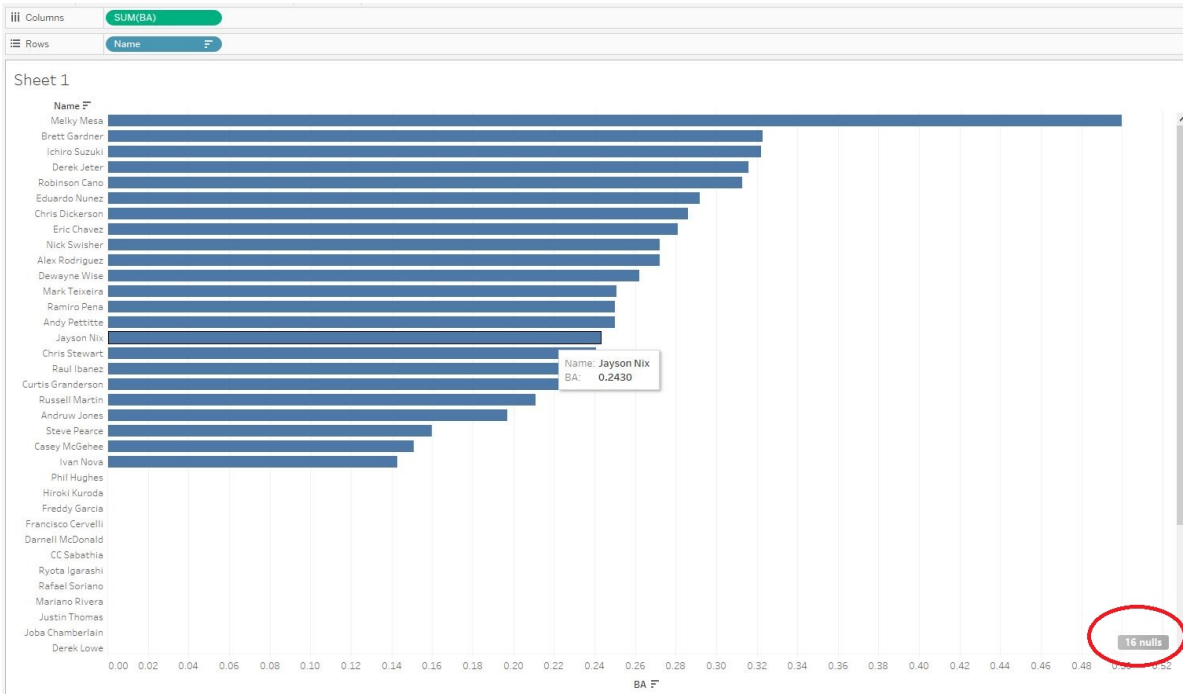
Drag 2012 Team Player Stats from the left-hand to the center of canvas and drop it. Check that your data has been loaded correctly.



After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

Step 2. Create a bar chart showing batting average

We first drag and drop Name from the dimension area to the Rows shelf, which lists all the players. We then drag and drop BA (batting average) from the measures area onto the Columns shelf, which creates a bar chart. We sort the bar chart in descending order.

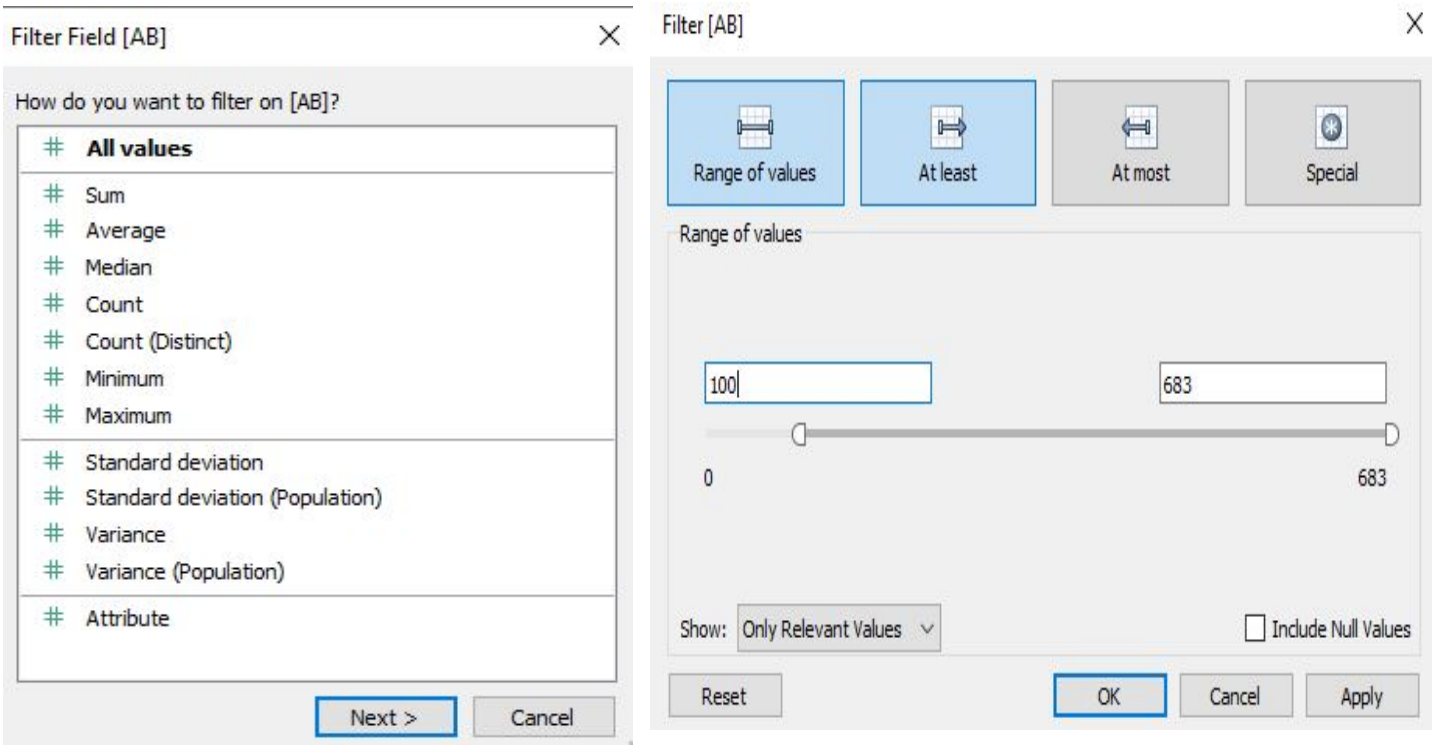


From the bar chart, we can see that almost half of the players have a batting average of 0.00. In the lower-right corner of the view, a note of 16 nulls indicates that 16 players didn't have any at bats.

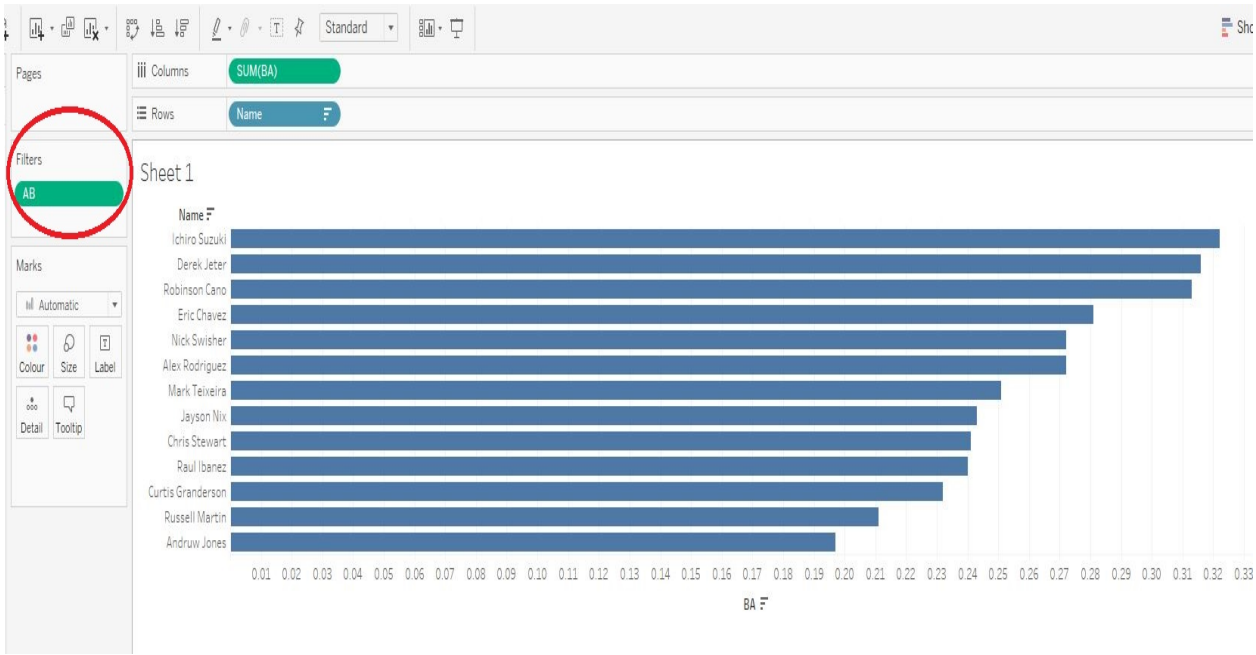
Step 3. Use filters to restrict certain data

We can restrict certain rows of data to be part of visualization. For example, from the above bar chart, we can see that almost half of the players have a batting average of 0.00. So, we can filter out players who didn't have a certain number of at bats.
To achieve this, pull a dimension or a measure onto the Filters card.

Drag AB (for at bats) field from Data pane to the Filers shelf above the Marks card, a dialog box will be opened:



We want to filter out players who have less than 100 at bats, so put 100 in the box. We can also drag the slider to the left or right to set the minimum number of at bats. After clicking Ok, the view of the filtered BA bar chart is shown:



Step 3. Place a quick filter to the view

We can add a Quick Filter to the view so that it is easy to set the minimum number of the field to be filtered out.

Click the small down arrow on AB filed in Filters and select Show Filter.

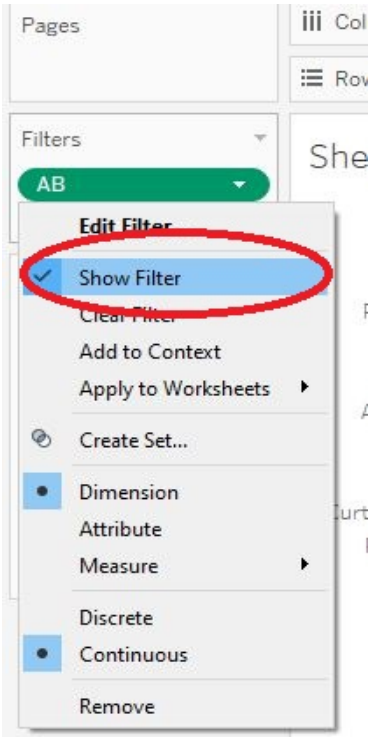
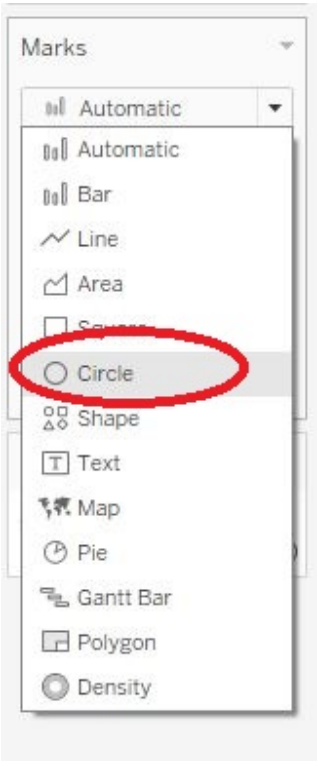
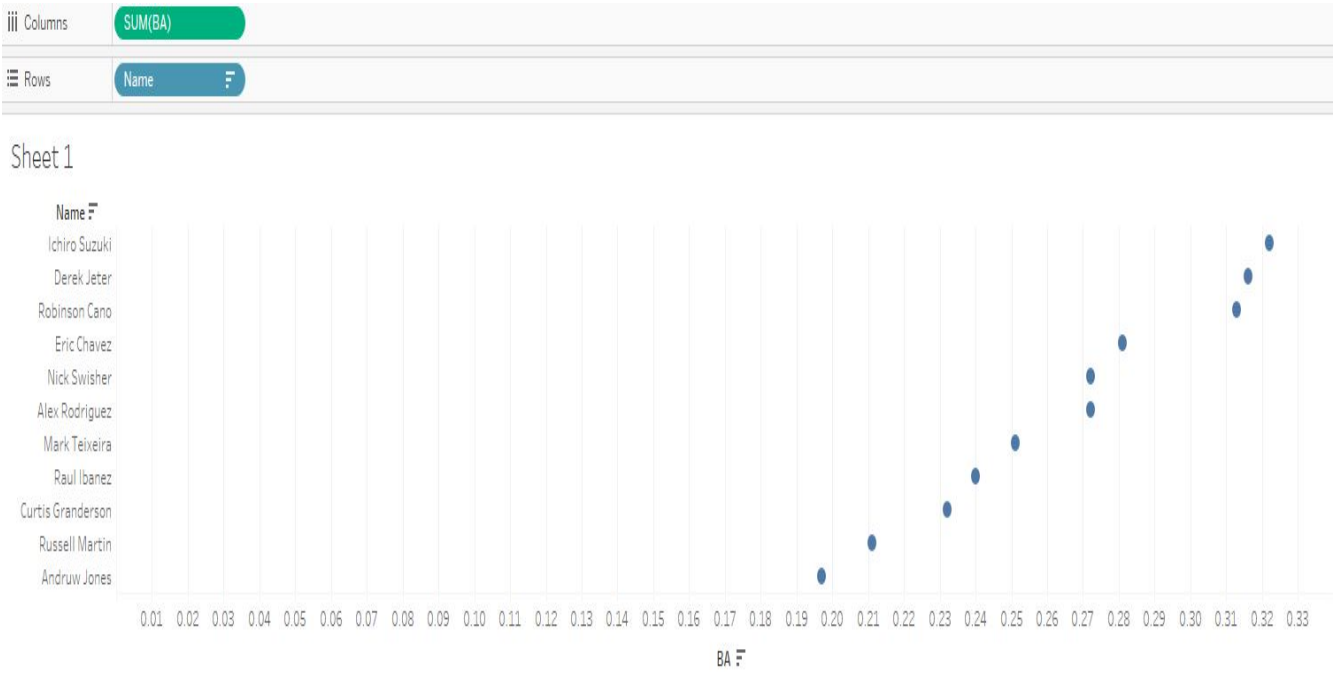


Tableau will place a slider control on the view. You can move the slider control to a proper location.



We then change the bar chart to dot chart:
Select Circle mark from the Marks dropdown in the Marks card.



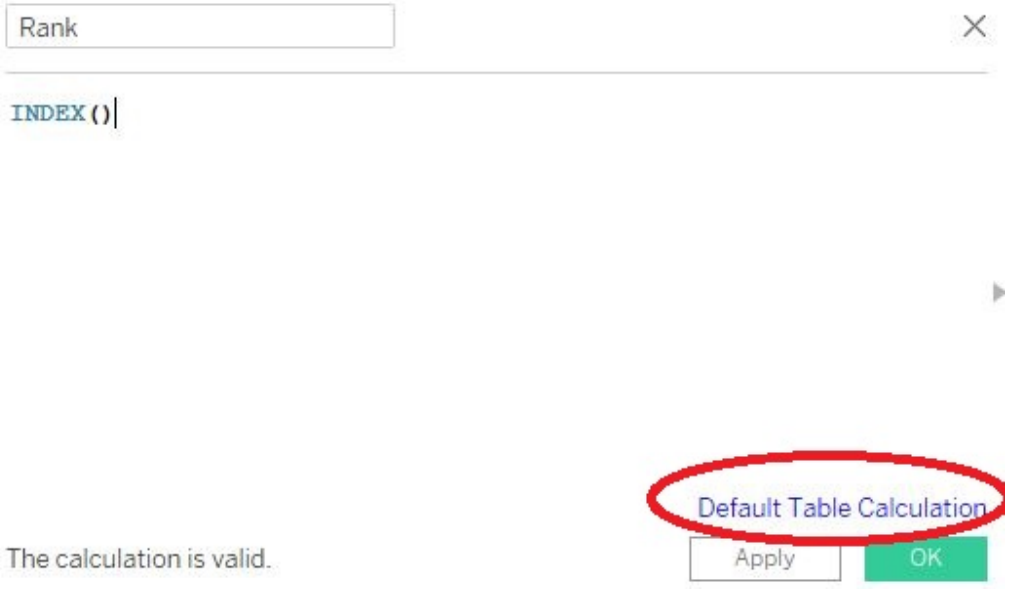


Step 4. Add a rank to the view

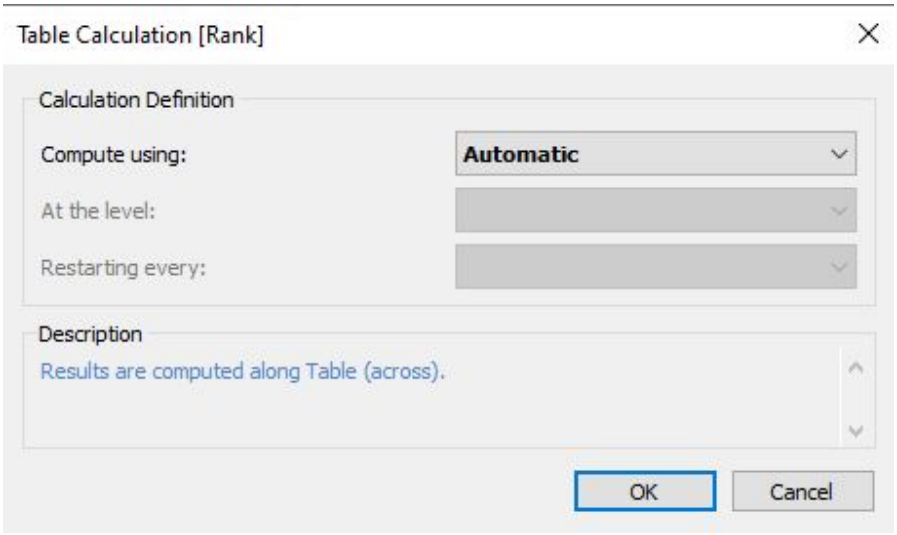
Now we add a rank to the players in order to quickly pick out the order of players from 1 to 11.

Right click any location in Dimensions or Measures area in Data pane and select Create Calculated Field.

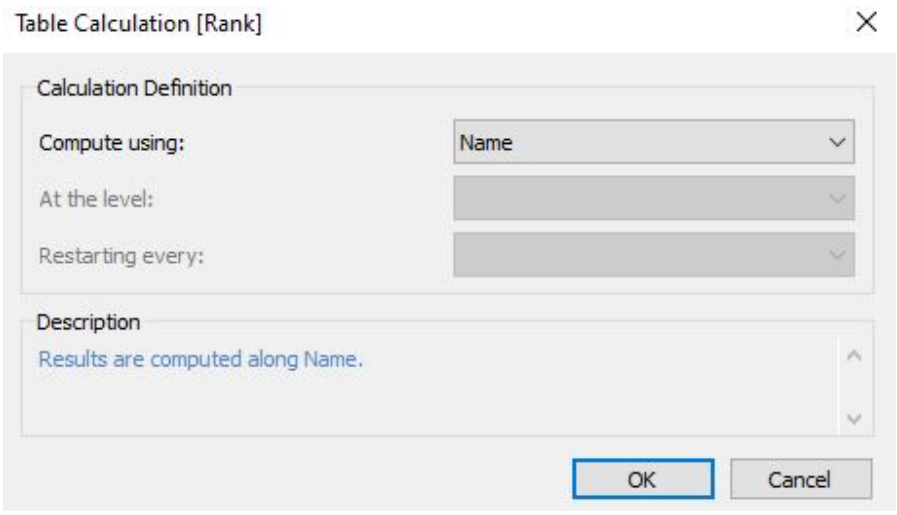
Then enter the name of the calculated field and enter code in code editor area to perform the Rank calculation as following.



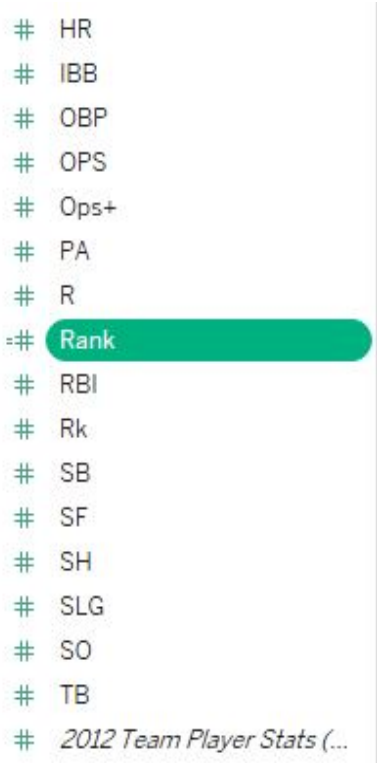
"The calculation is valid" and blue text "Default Table Calculation" appear in the box. Then click on this blue text, we get



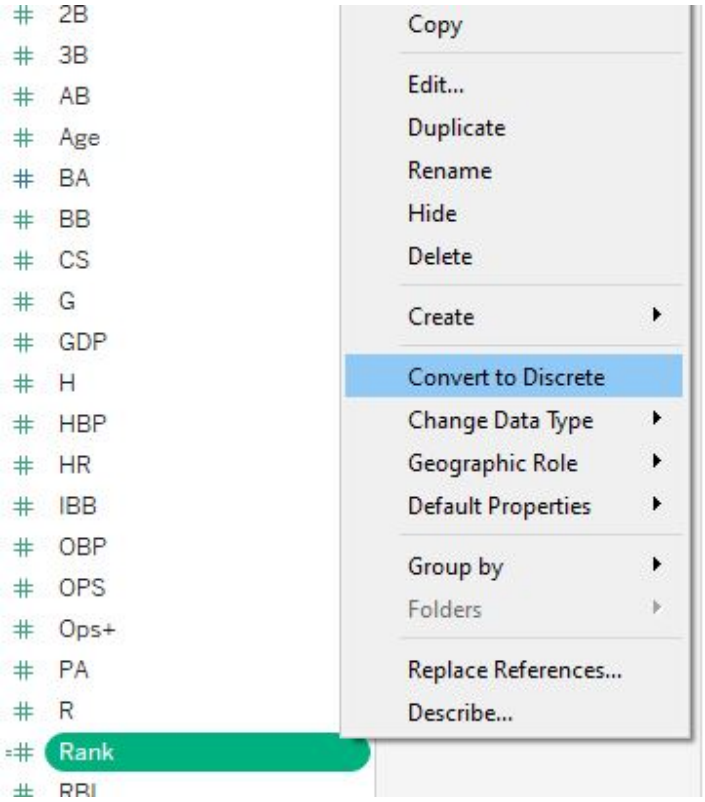
In the box of Compute using, change Automatic to Name, which indicates that the rank will apply to the Name.



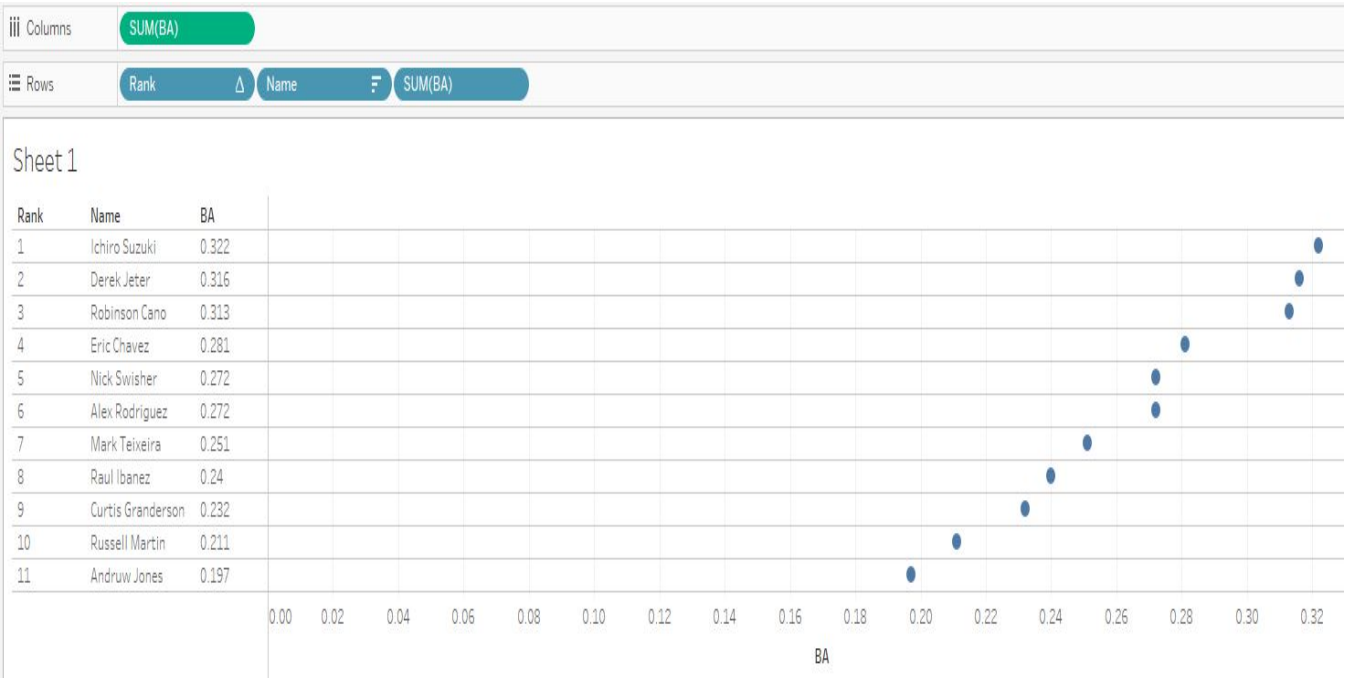
After clicking OK button, a new field of Rank you just created appears in the Measures area.



After right clicking Rank and selecting Convert to Discrete, Rank will be changed to blue color.



Finally, to add the Rank field to the bar chart, just drag it to the left of Name on the Rows shelf. We can also drag BA from Data pane to the Rows shelf.



Project 3: Table calculations

Table Calculations allow you to create computations that span across different marks in your visualization. Predefined Quick Table Calculations make it possible to quickly include several commonly used calculations without having to type any formulas.

Table Calculations are derived from the data shown in the view. Once you have built a first chart, you can quickly add a Table Calculation by using one of the many predefined Quick Table Calculations, including **proportions, rankings, cumulative sums, and moving averages.**

Task: The given dataset provides statistic information on baseball game's players, including players' names, team batting, team pitching, and batting average, etc. The objective of this task is to compute percent of total.

1. Download Dataset: baseball-player-stats.xlsx from Week 7 in Moodle

Step 1. Load the data into Tableau

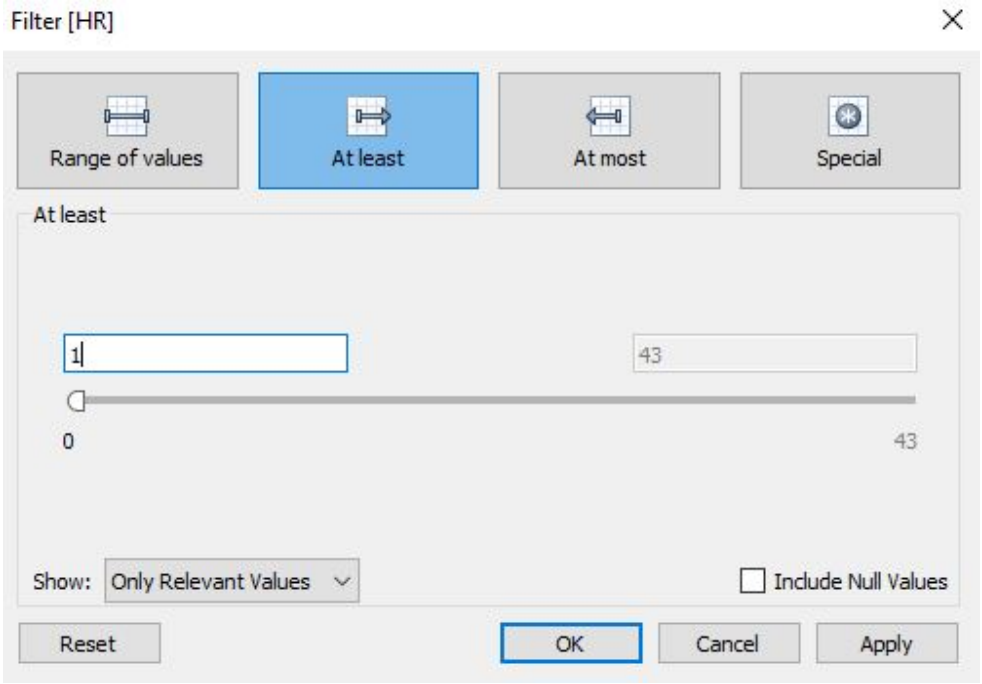
Drag 2012 Team Player Stats from the left-hand to the center of canvas and drop it. Check that your data has been loaded correctly.

After clicking Sheet 1 tap on the bottom, the data will be connected to your computer and shown in the data pane on.

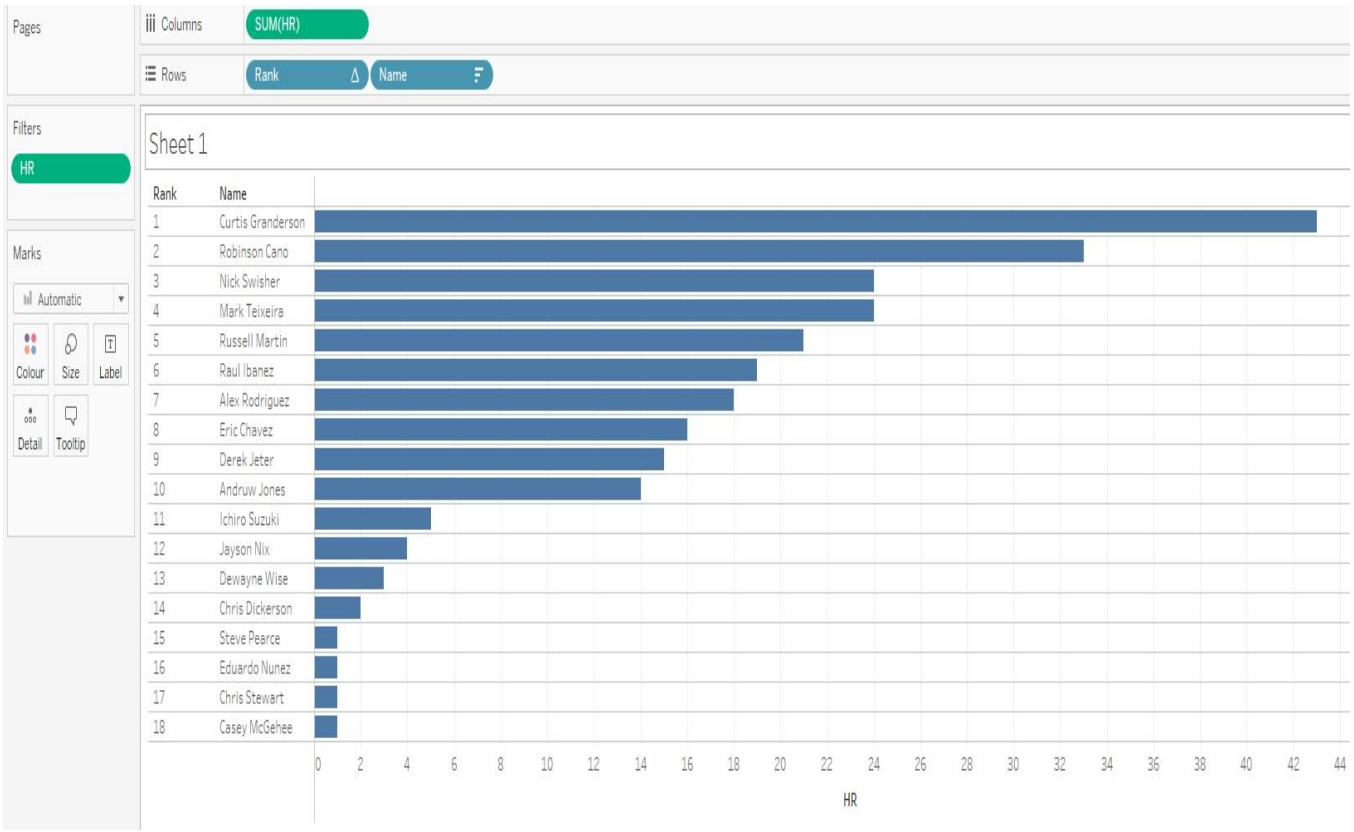
Step 2. Create a bar chart showing at least one home run

We first drag and drop Name from the dimension area to the Rows shelf, which lists all the players. We then drag and drop HR (for home runs) from the measures area onto the Columns shelf, which creates a bar chart. We sort the bar chart in descending order.

Then drag HR to the Filter shelf and selecting At Least one home run:



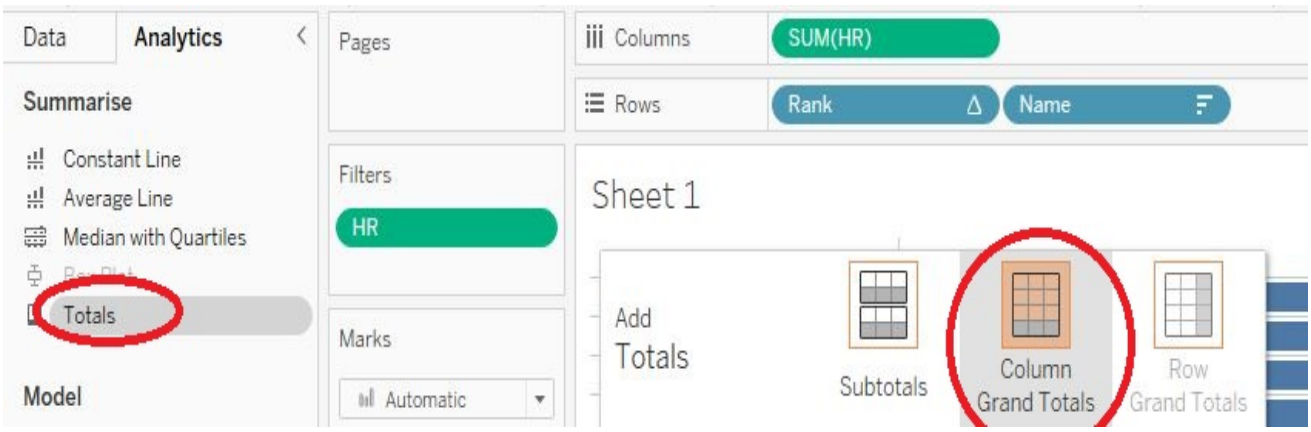
We then add a Rank field as we did as before

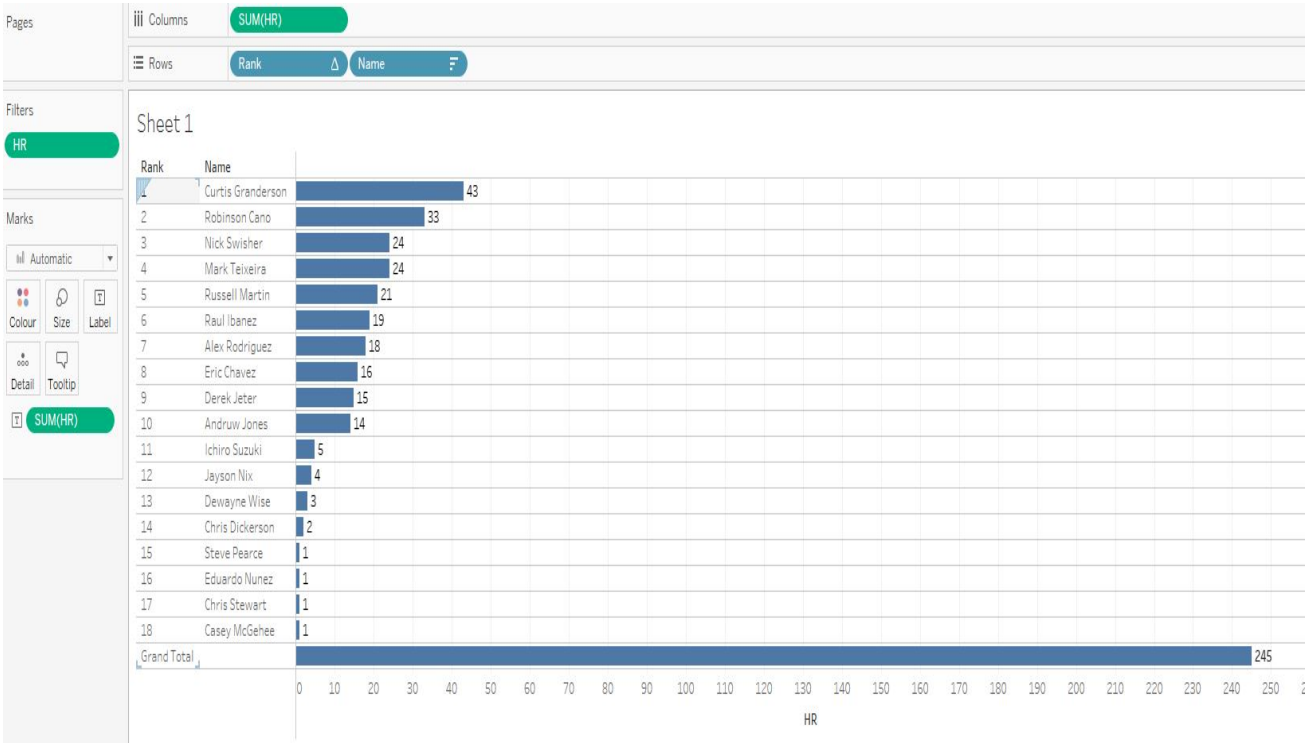


Step 3. Add team total to the view

We want to compare players' home run to overall team number.

Select Analytics menu from the top, and then click Totals and move it and select Column Grand Totals. Drag HR to the Label shelf in Marks card.

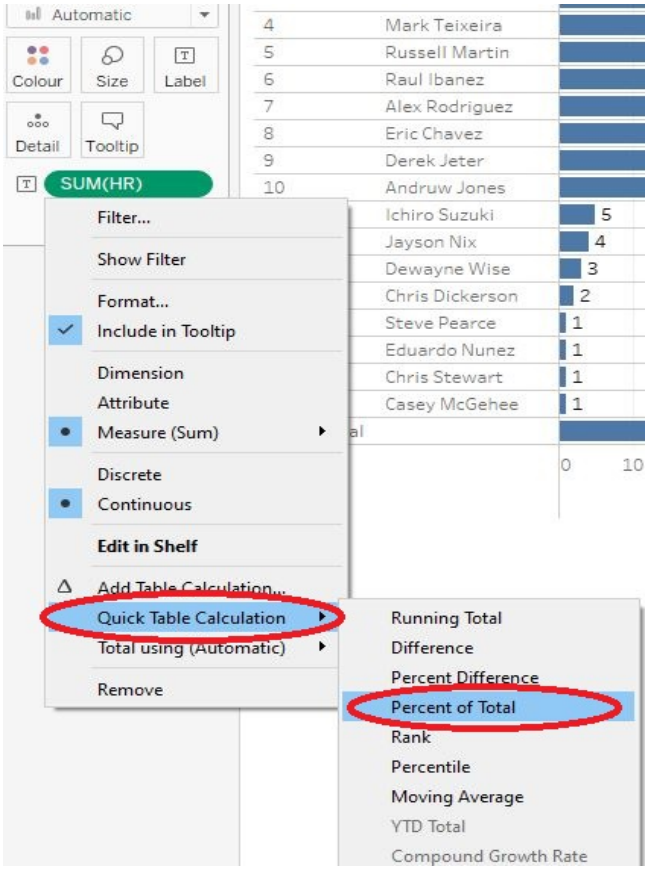


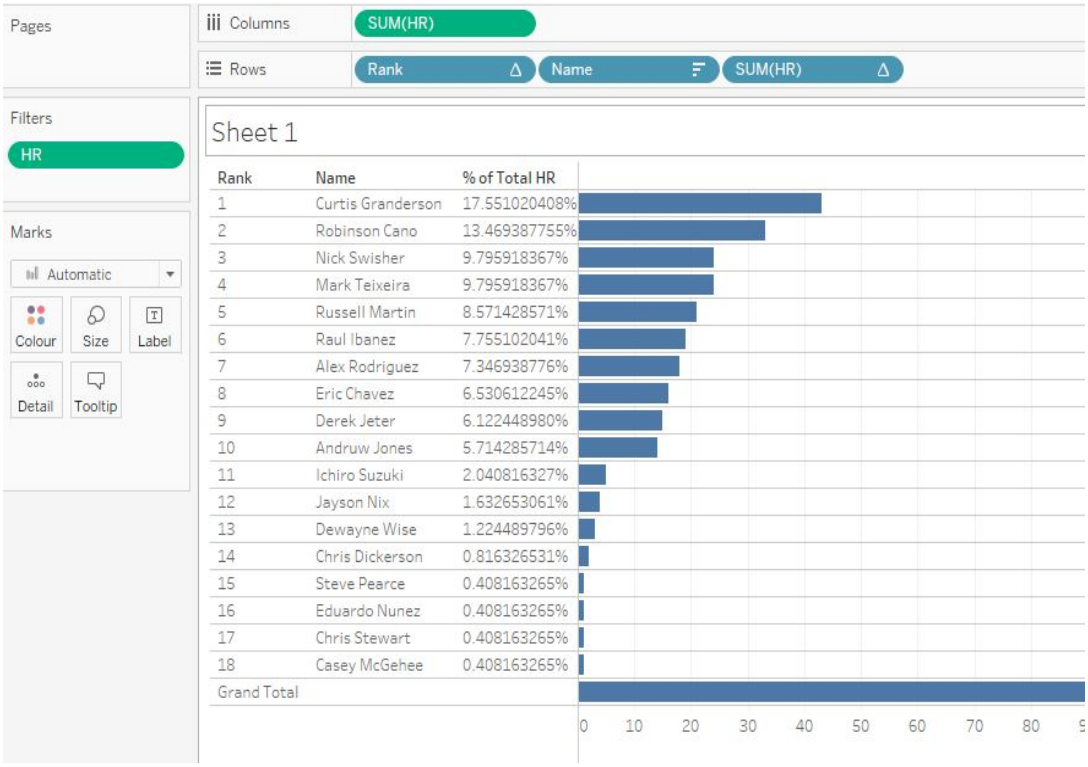


From this view, we can see that the grand total of team hitting home run is 245, whiles Cano hits 33 home runs.

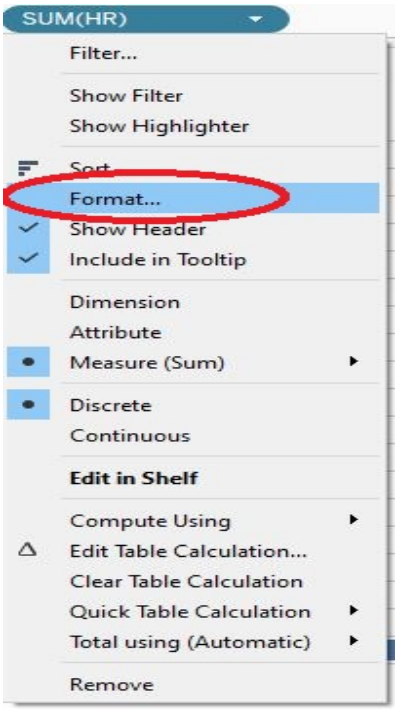
Step 4. Show percent of total in the view

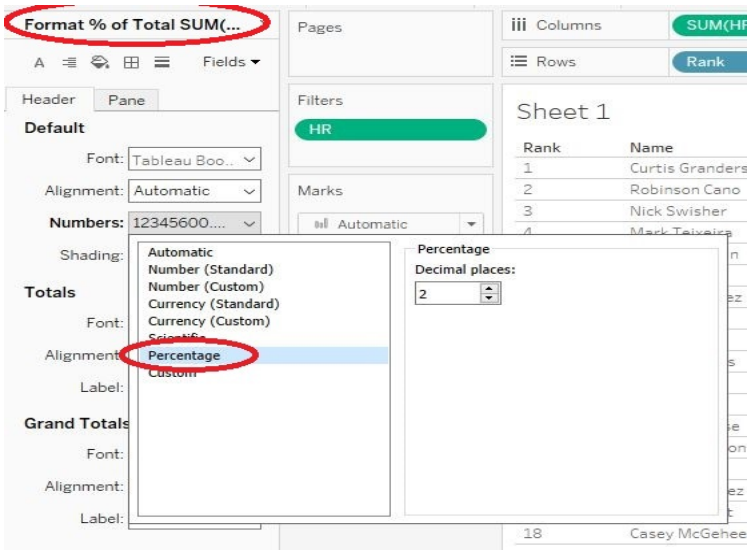
To add a percent of total to each player bar, just click the small down arrow on SUM(HR) in Marks card and select Quick Table Calculation | Percent of Total


Automatic
Colour
Size
Label
Detail
Tooltip
SUM(HR)
Filter...
Show Filter
Format...
Include in Tooltip
Dimension
Attribute
Measure (Sum)
Discrete
Continuous
Edit in Shelf
Add Table Calculation...
Quick Table Calculation
Total using (Automatic)
Remove
Running Total
Difference
Percent Difference
Percent of Total
Rank
Percentile
Moving Average
YTD Total
Compound Growth Rate
4 Mark Teixeira
5 Russell Martin
6 Raul Ibanez
7 Alex Rodriguez
8 Eric Chavez
9 Derek Jeter
10 Andruw Jones
Ichiro Suzuki 5
Jayson Nix 4
Dewayne Wise 3
Chris Dickerson 2
Steve Pearce 1
Eduardo Nunez 1
Chris Stewart 1
Casey McGehee 1
0
10

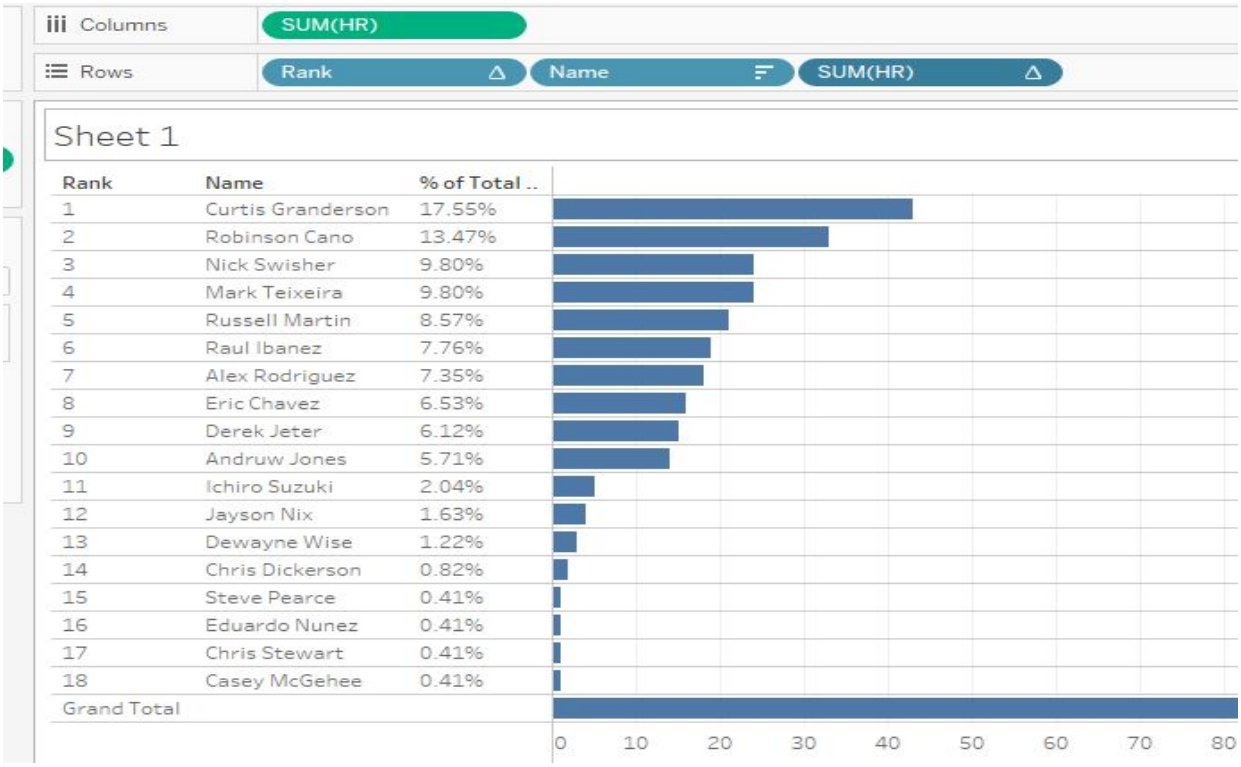


To change the number of digits shown in % of Total HR, click SUM(HR) and select Format





After selecting percentage, we get:



Project 4. Create a Professional Report

Summarize the above experiments procedure, results, answering for questions and screenshots (project 1, 2, 3) into one report. Your report is the assignment that is required to be submitted for evaluation on week 8. Create a report by following the below steps.

Please note that

- as you may have a different version of Microsoft Word, the **following steps are just for your references.**

You can add a chapter called Chapter 7 in your previous report.

1. Open your last week's report and find the end of last week's report.
2. Copy this week's related experimental results, your findings and Screenshots, and paste them at the end of last week's report.
3. Delete original Table of Content you created.
4. Select all content , align all text to both left and right margin
5. Use shortcut key approach, generate Chapter 7: **Complex data visualizations (2)**
6. Then use shortcut key approach to generate proper sub-chapters for this week's lab work.
7. Insert Table of Contents to your report.